

Database Design for Airbnb-like Platform

Abstract:

The airbnb_company database is designed to effectively manage an Airbnb-like platform, incorporating 21 interrelated tables. The schema is highly organized and covers critical platform functions, such as user administration, property management, booking transactions, payment processing, reviews, and more. At the core of the database lies the user table, which plays a central role, referring to both hosts and guests. The Host and Guest tables store additional details like bank. account and payment options.

A structured approach guided the design process through multiple stages.

1. Requirements gathering: After identifying key entities system requirements were collected for each entity together with their relationships before defining the relevant constraints.
2. The team constructed an entity relationship diagram to illustrate the database's high-level structure during the conceptual design phase.
3. The logical design phase transformed conceptual design elements into a structured logical schema. Redundancy reduction and data integrity enhancement were achieved through the normalization process of the schema.
4. MySQL was used to develop the physical database schema. The schema outlined the entities, attributes, data types, constraints, and indexes for each table. mock data was added to the tables.

The Property table stores property data which connects with the Location table for address management.

the Amenity table shows all the facilities available on the property.

The Booking table maintains the relationship between guest and host and can provide the details information about check - in and check – out dates and status of booking.

The Payment table manages payment transactions between guests and properties associated with booking. Details such as amount, commission, payment date, and payment method are stored, facilitating secure and efficient payment processing.

The Review table helps the users, receive permissions to assign ratings and leave feedback about their experiences.

The location table provides geographical information about property based on the user's search for the specified country and cities.

The schema incorporates multiple supporting tables which work to improve platform functionality.

The platform schema supports its functionality through additional tables including Commission, Payment, Message, Social Media, Dispute, Notification, Income, and Language.

The tables monitor commissions, payments, messaging activity, social media engagement, and dispute management.

The language preferences of users are stored in the language table, facilitating multi language support for the platform.

The system uses resolutions for disputes and notifications alongside income tracking and language preferences to present a full platform management solution.

Conclusion

The database structure enables support for multiple user roles along with property listings as well as bookings and payment transactions.

The database architecture maintains data integrity by implementing well-defined relationships and constraints as it manages reviews along with other functionalities. This Scalable system integration produces an efficient way to handle complex interactions. The system manages complex interactions and data flows which are standard for an Airbnb-style platform.